Ensemble of tree-based models

Part 2

Boosting and gradient boosting

Boosting vs Gradient Boosting

Traditional Boosting

`sklearn.ensemble.AdaBoostClassifier`

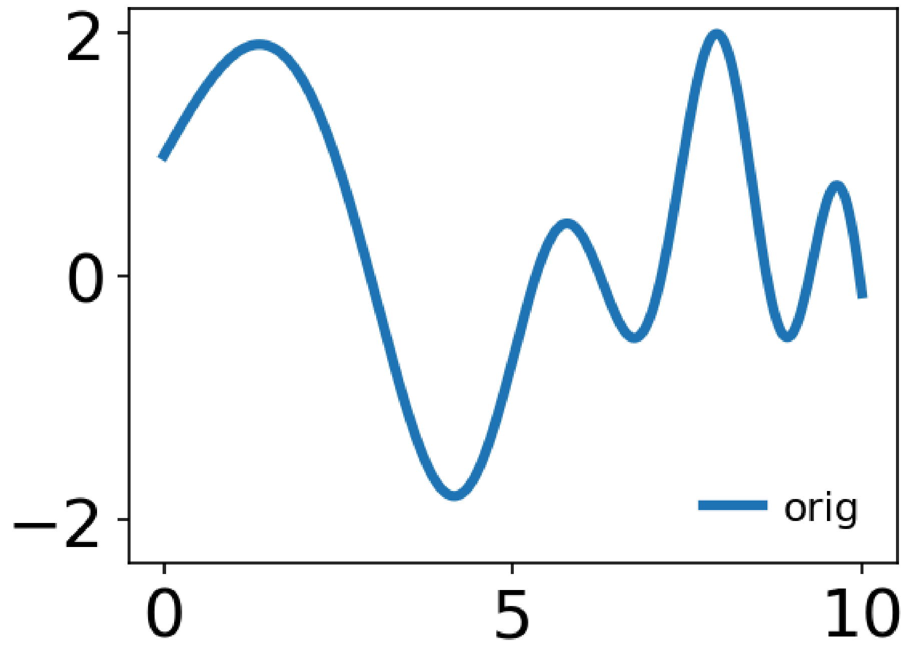
- Mispredicted samples are re-weighted at each step
- It is not recommended to use AdaBoost anymore

Gradient Boosting

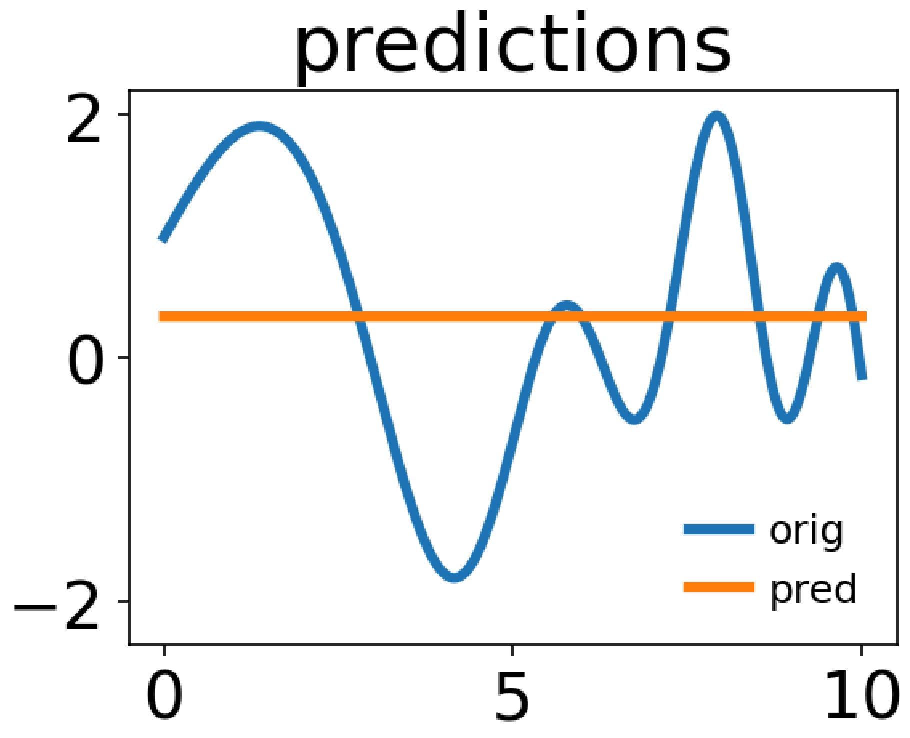
`sklearn.ensemble.HistGradientBoostingClassifier`

- Each new model in the sequence predicts the error of the previous models
- sklearn uses decision trees as the base model

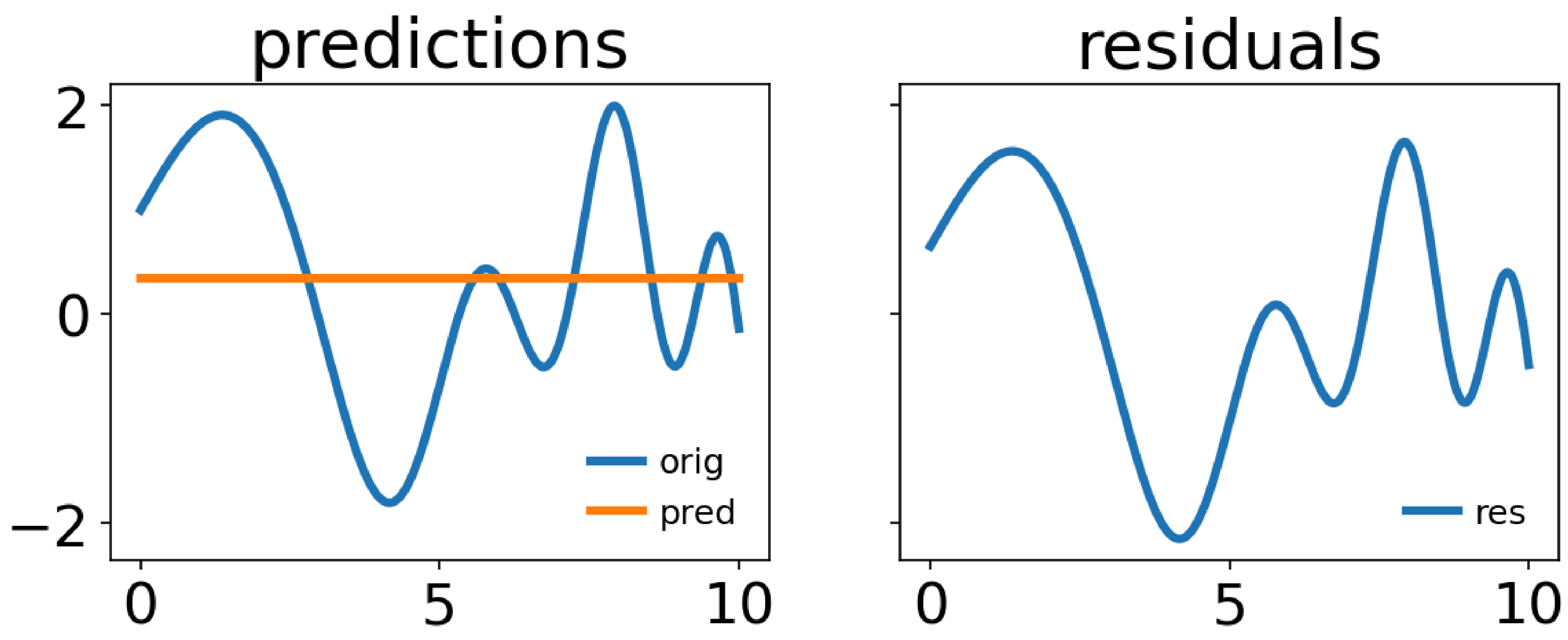
Gradient boosting for regression



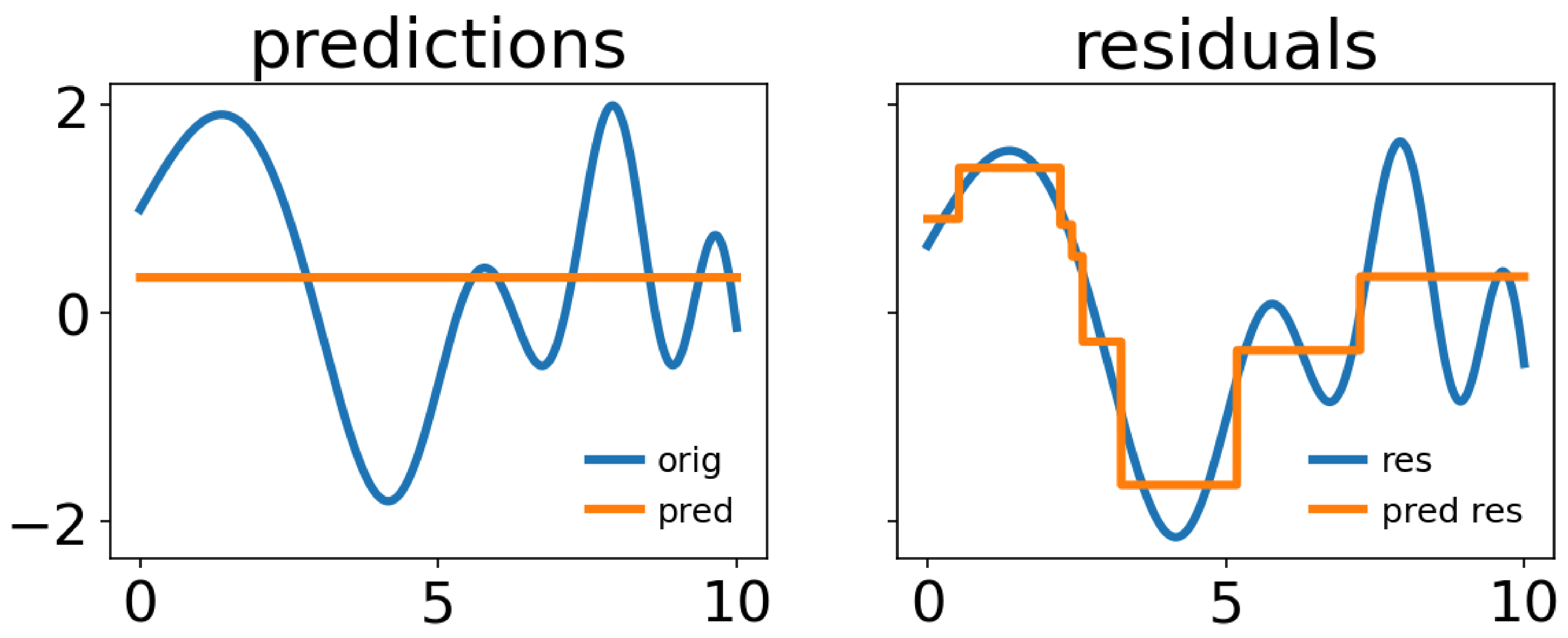
Gradient boosting for regression



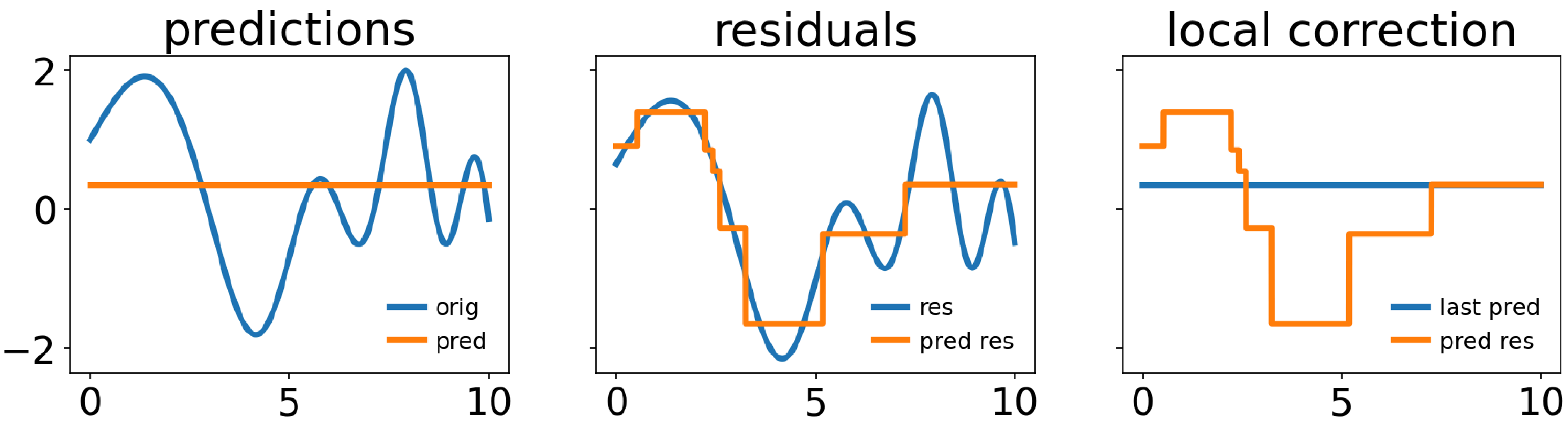
Gradient boosting for regression



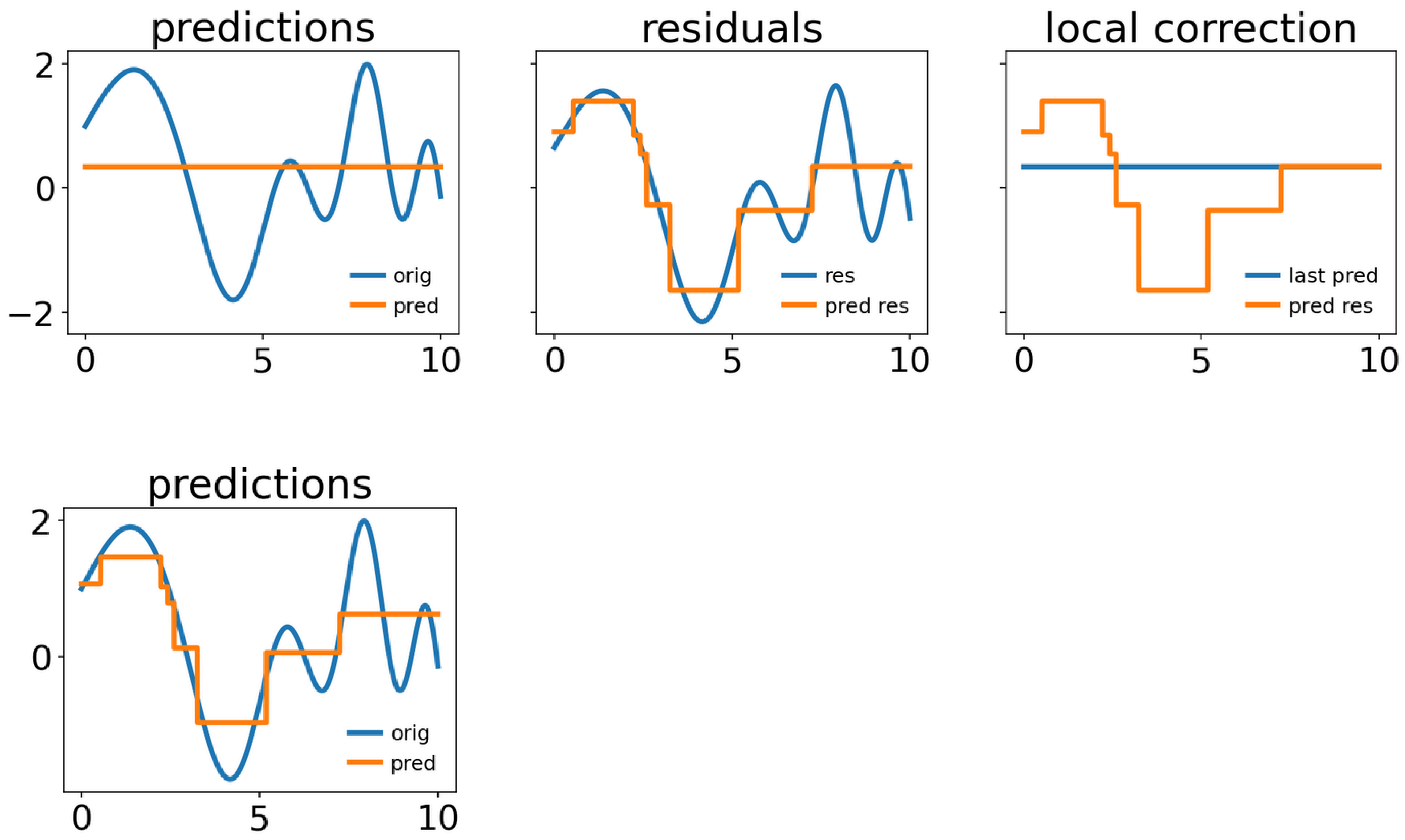
Gradient boosting for regression



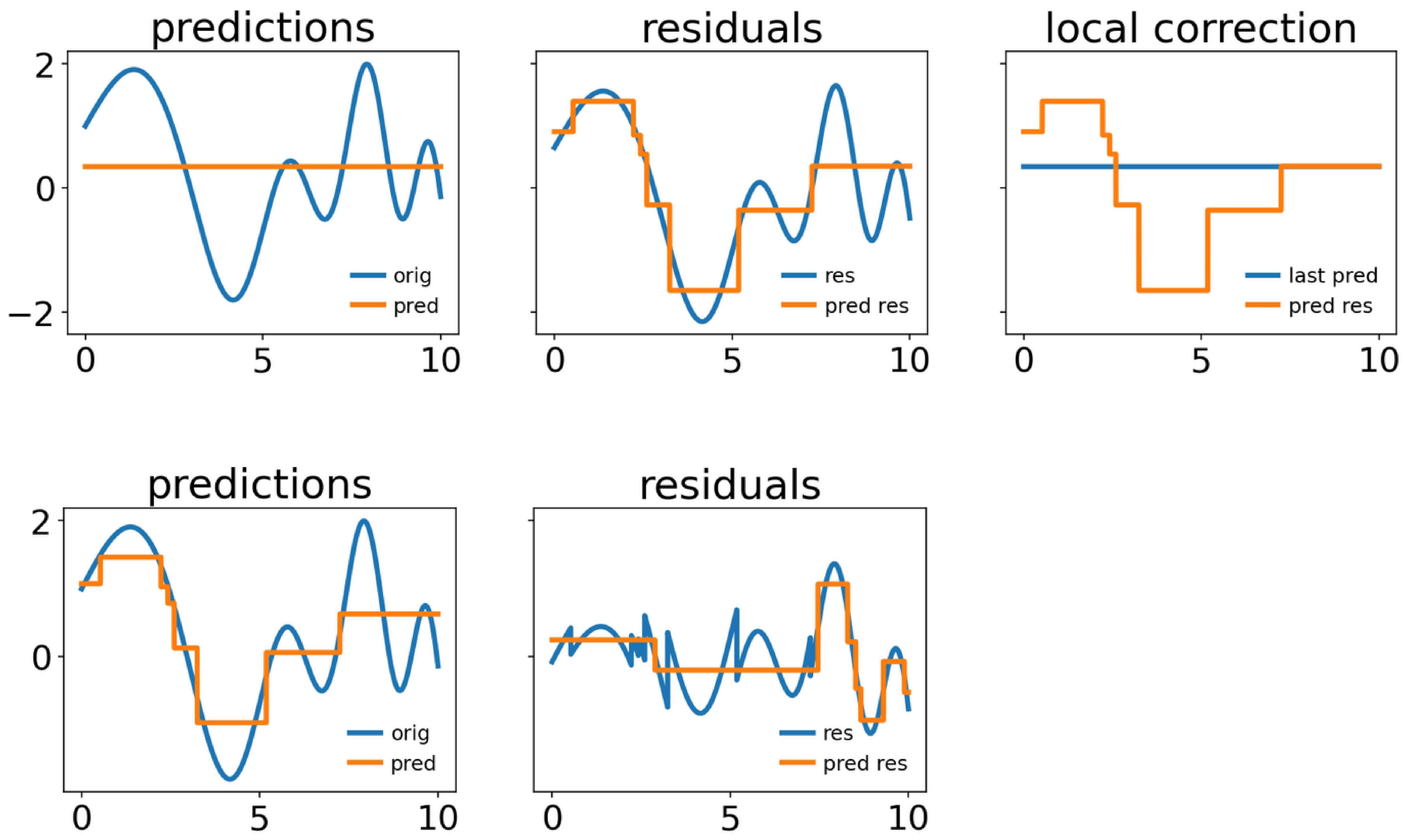
Gradient boosting for regression



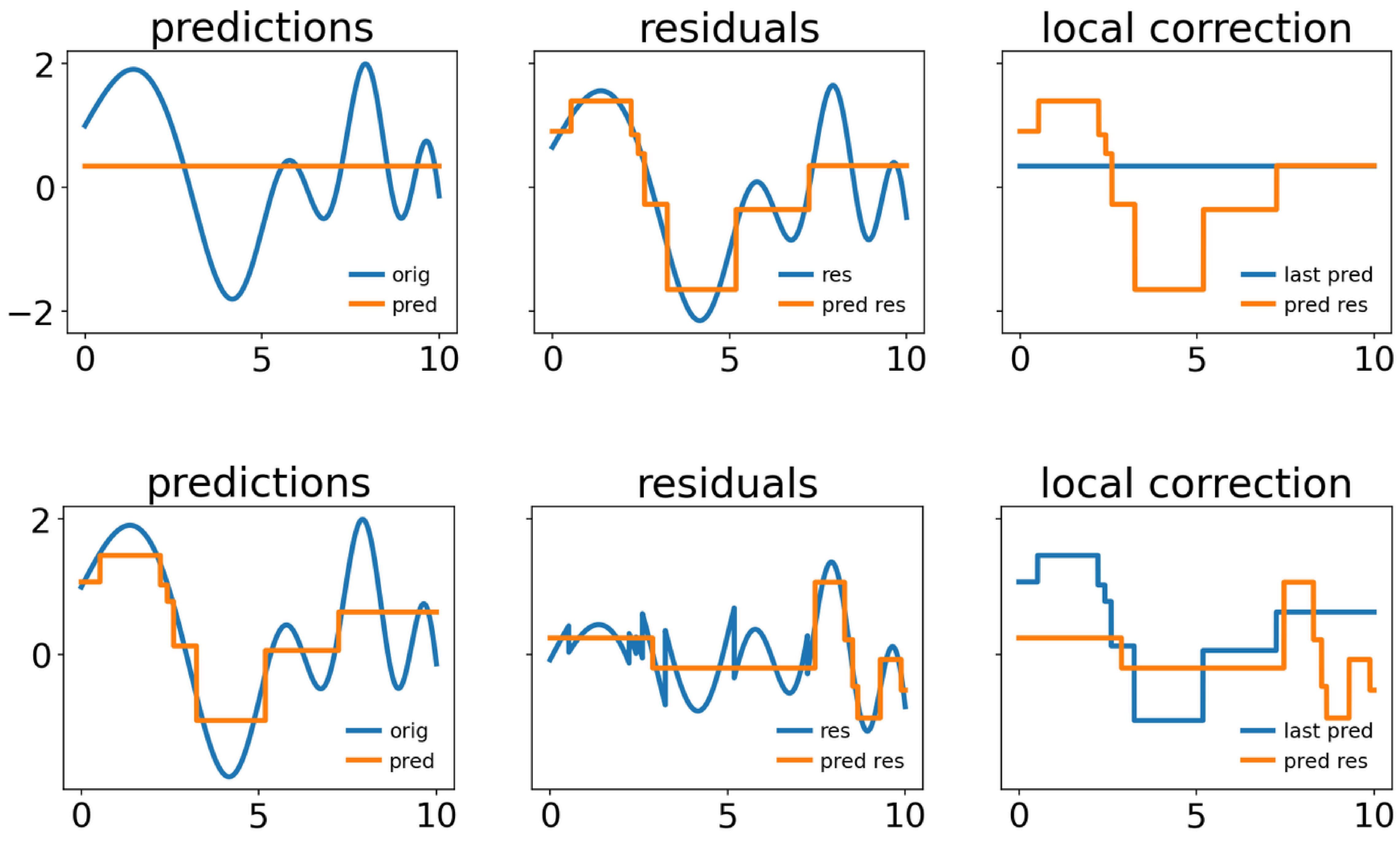
Gradient boosting for regression



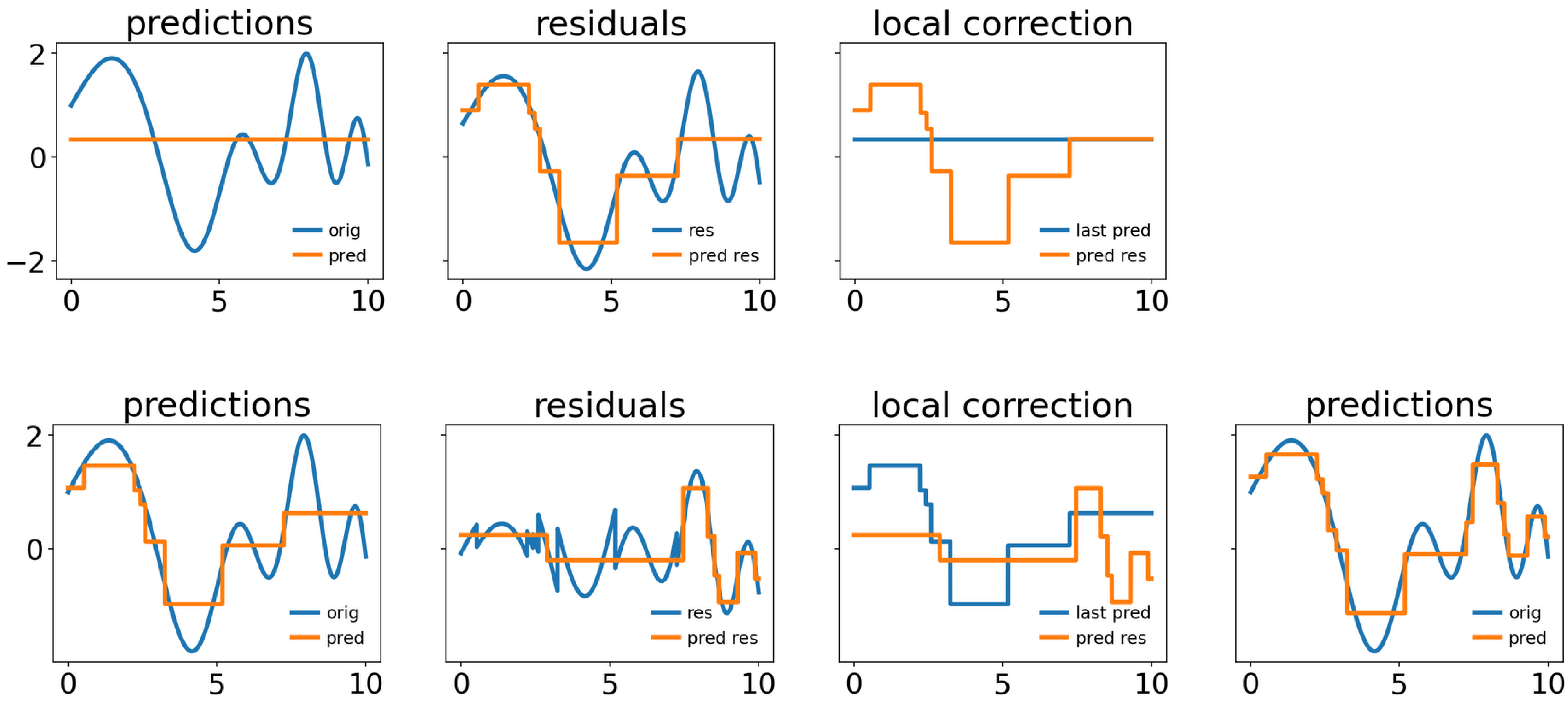
Gradient boosting for regression



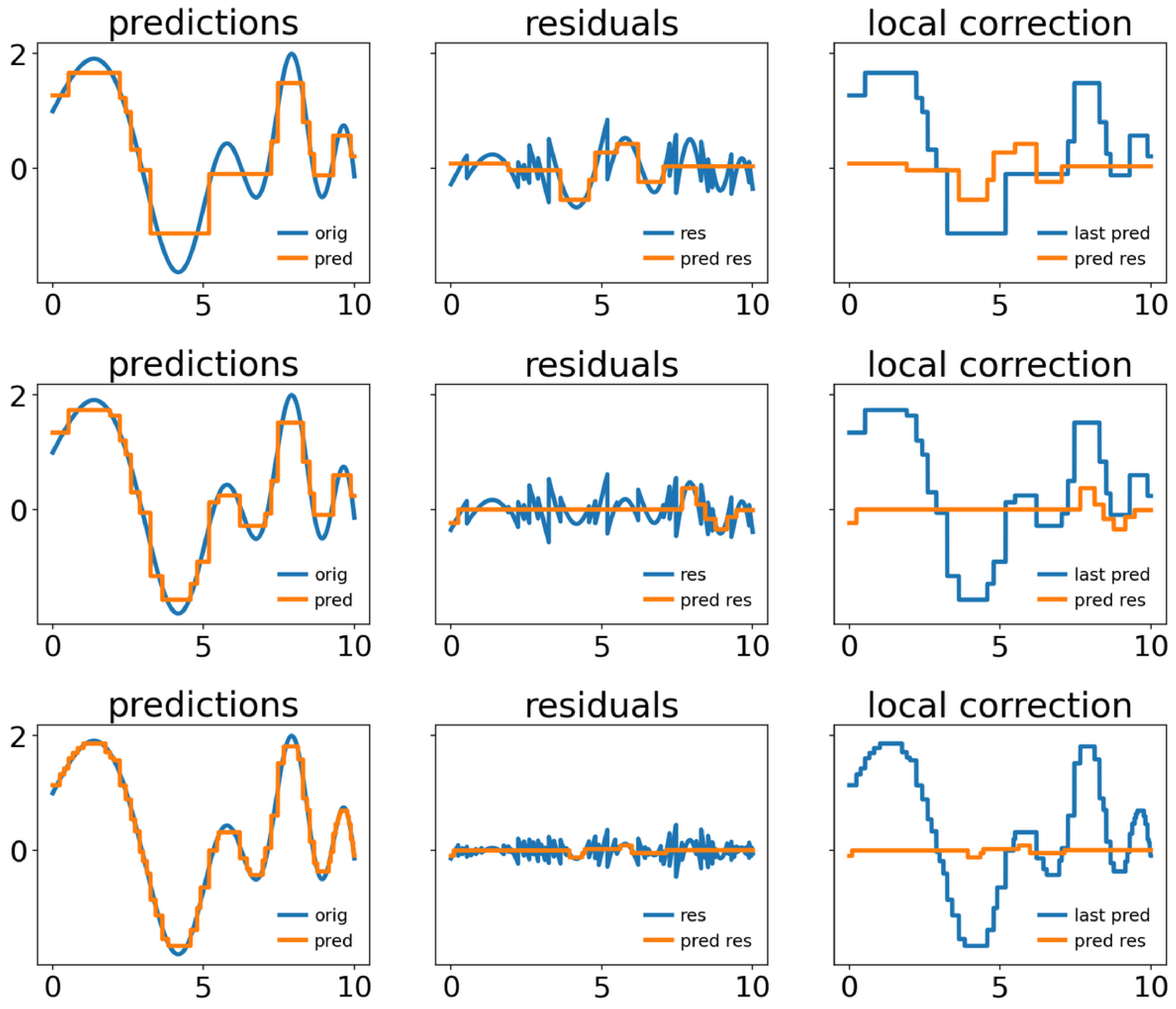
Gradient boosting for regression



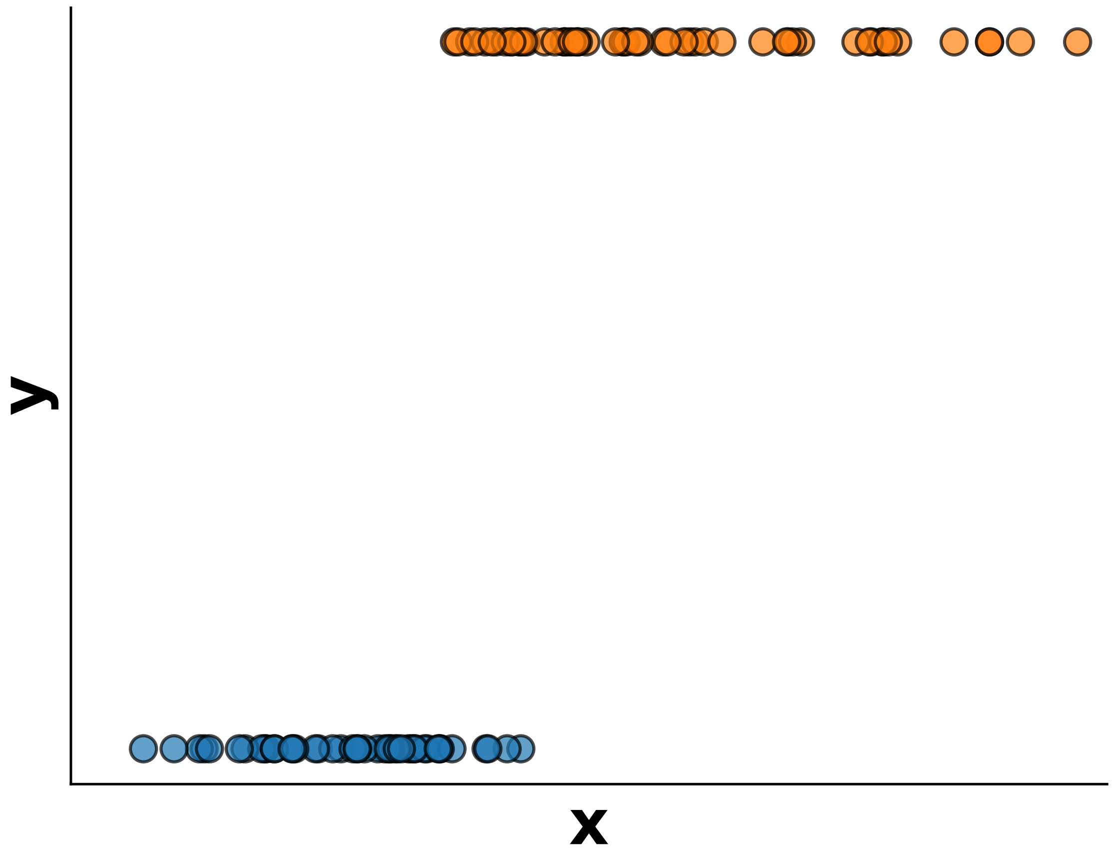
Gradient boosting for regression



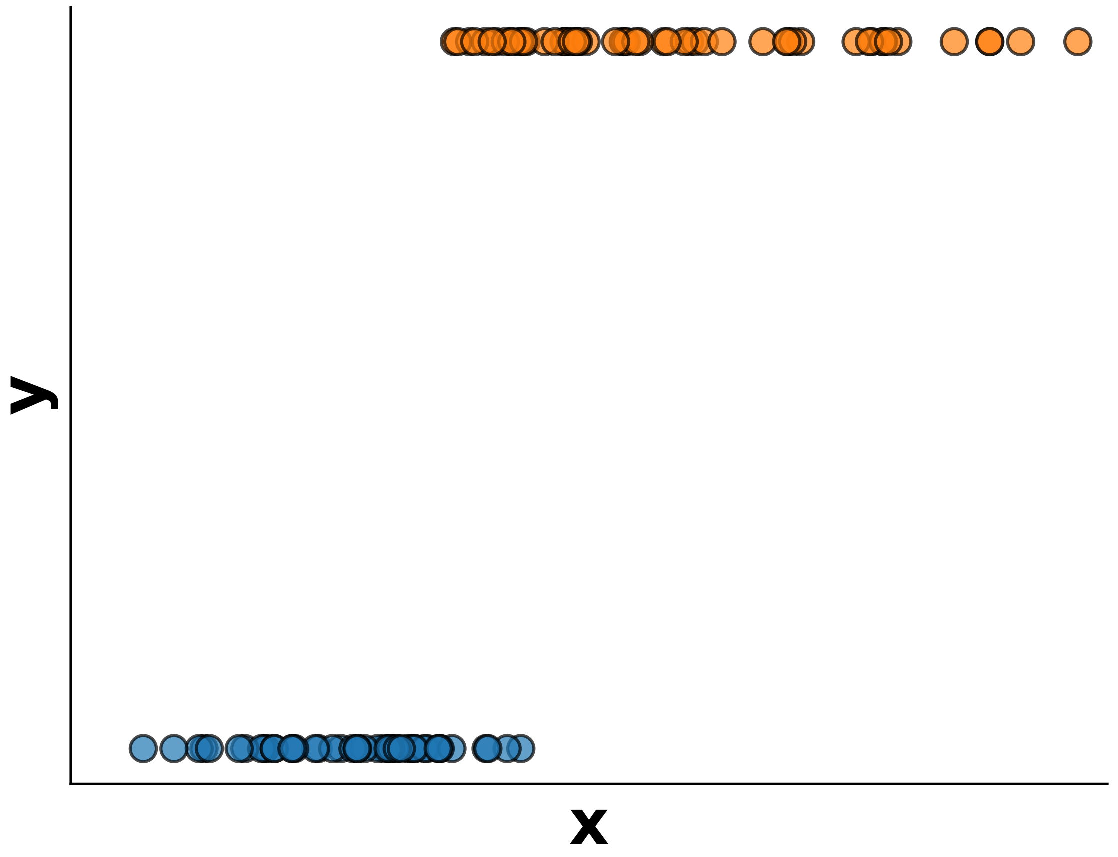
Gradient boosting for regression

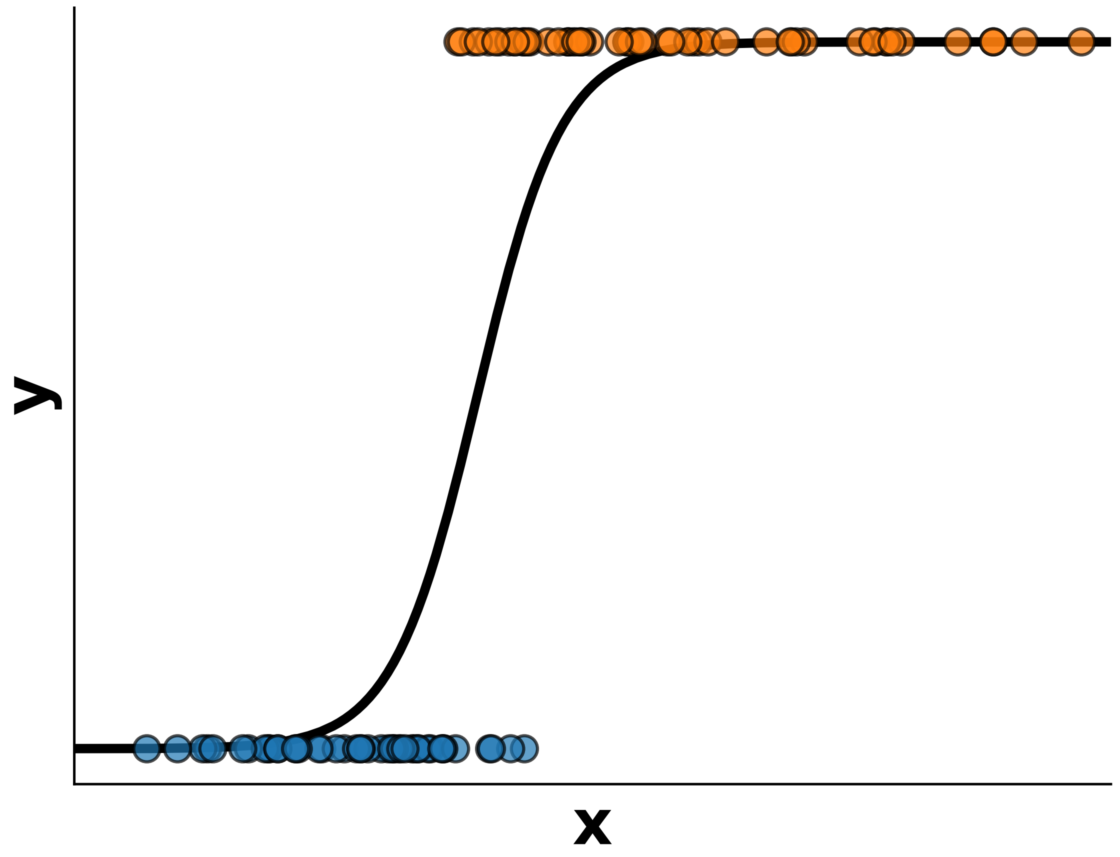


Gradient boosting for classification

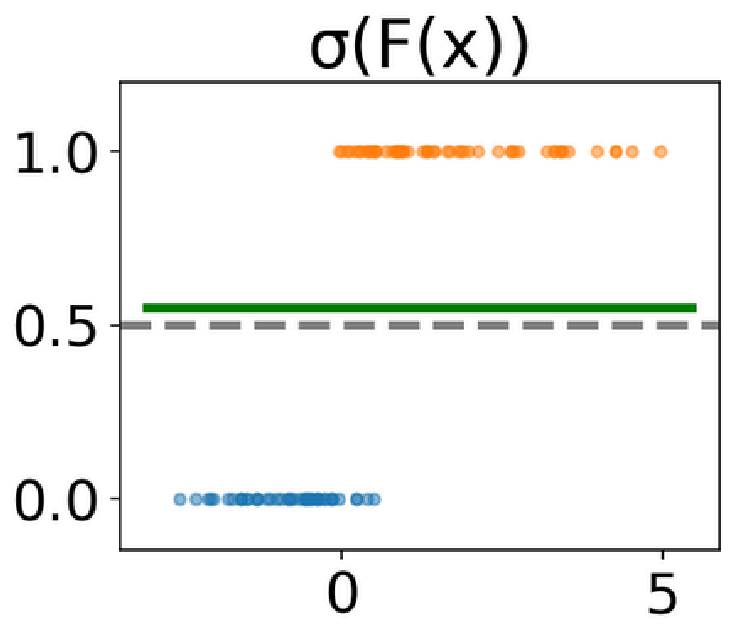


Gradient boosting for classification



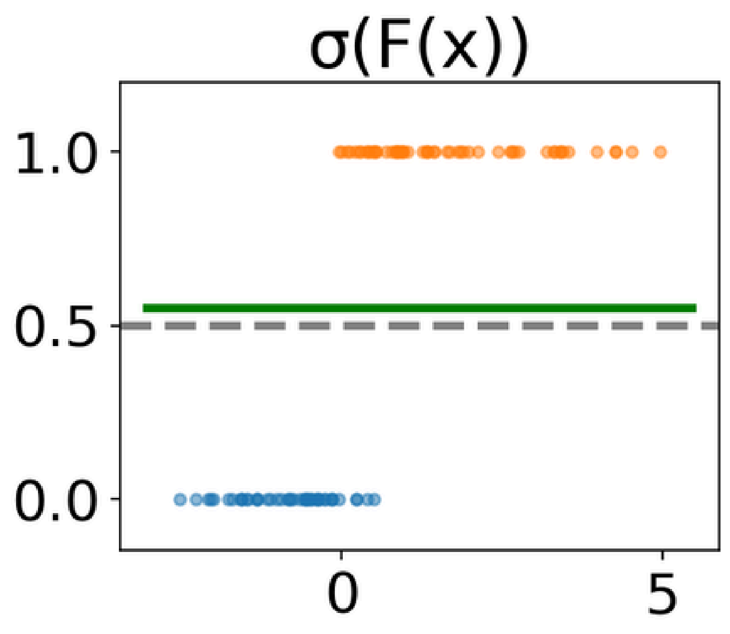


Gradient boosting for classification



$$\sigma(\ F(x)\ ) = \frac{1}{1 + e^{-\ F(x)}}$$

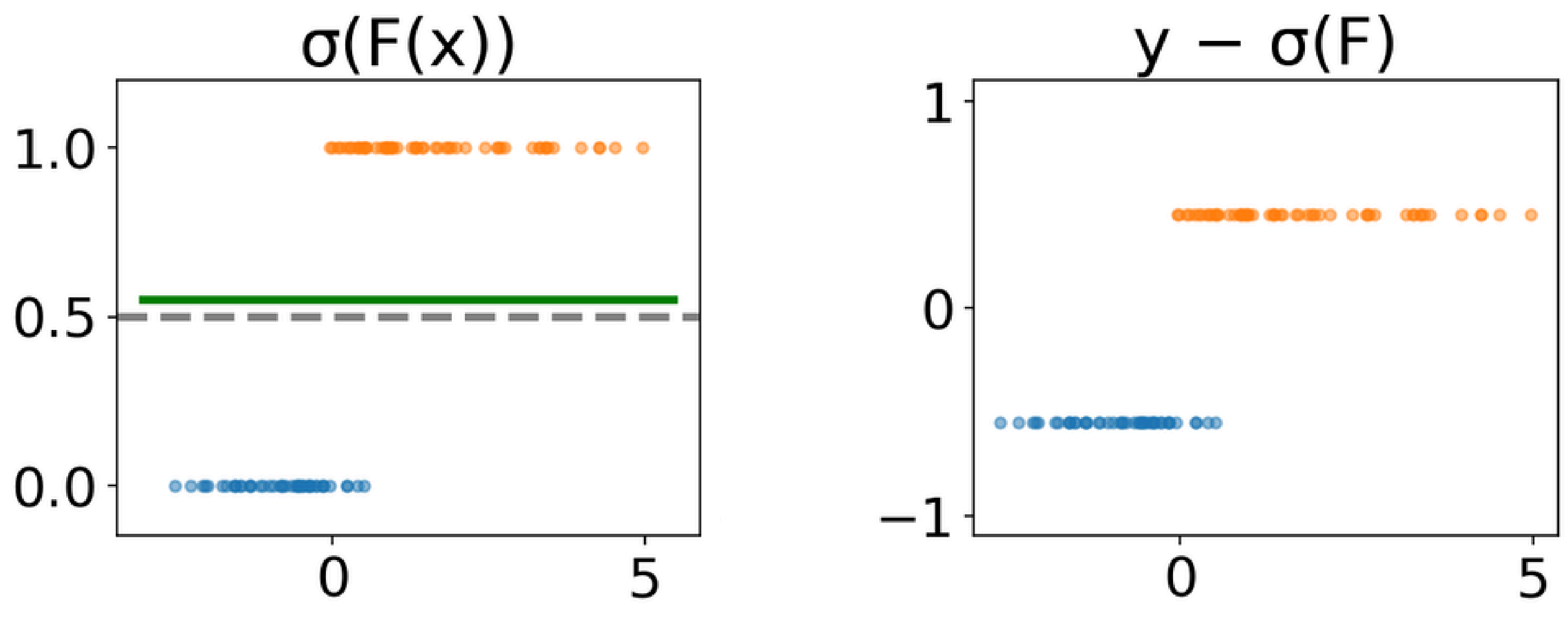
Gradient boosting for classification



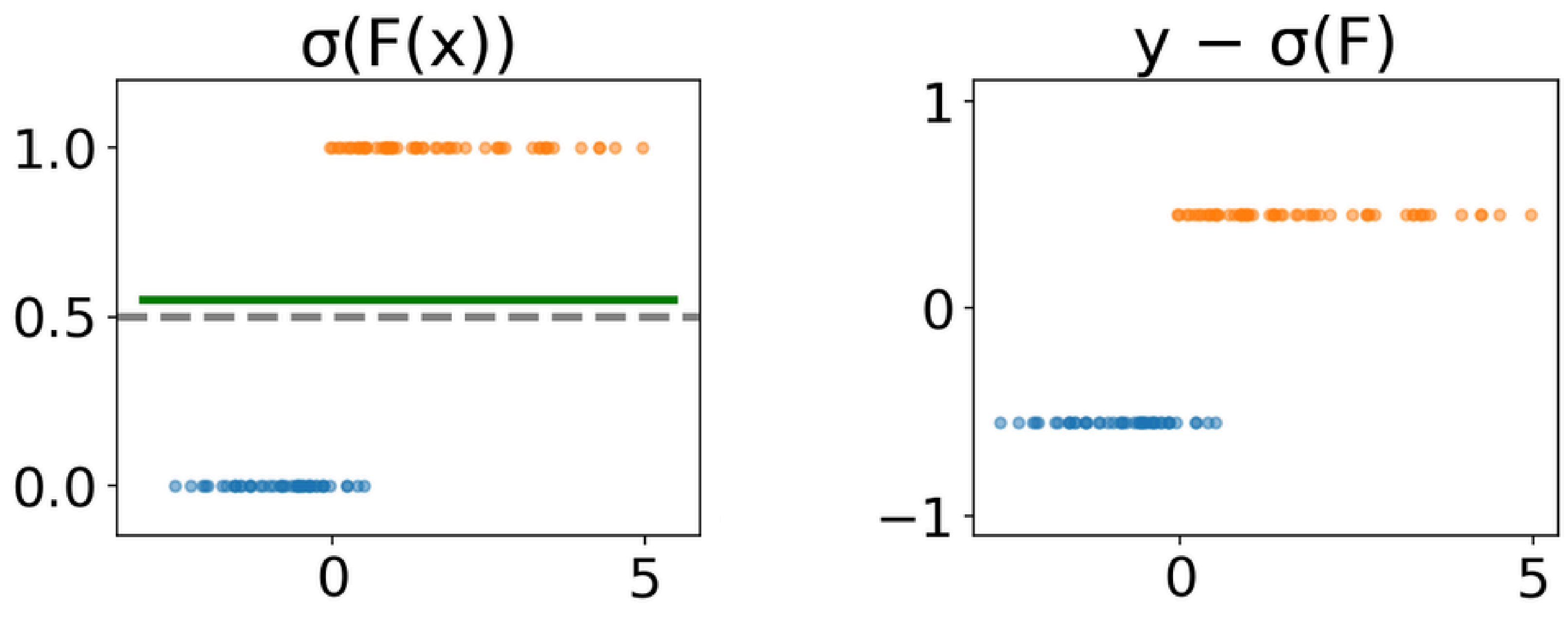
$$\sigma(\,F(x)\,) = \frac{1}{1 + e^{-F(x)}}$$

Note: The sigmoid of a constant is another constant

Gradient boosting for classification

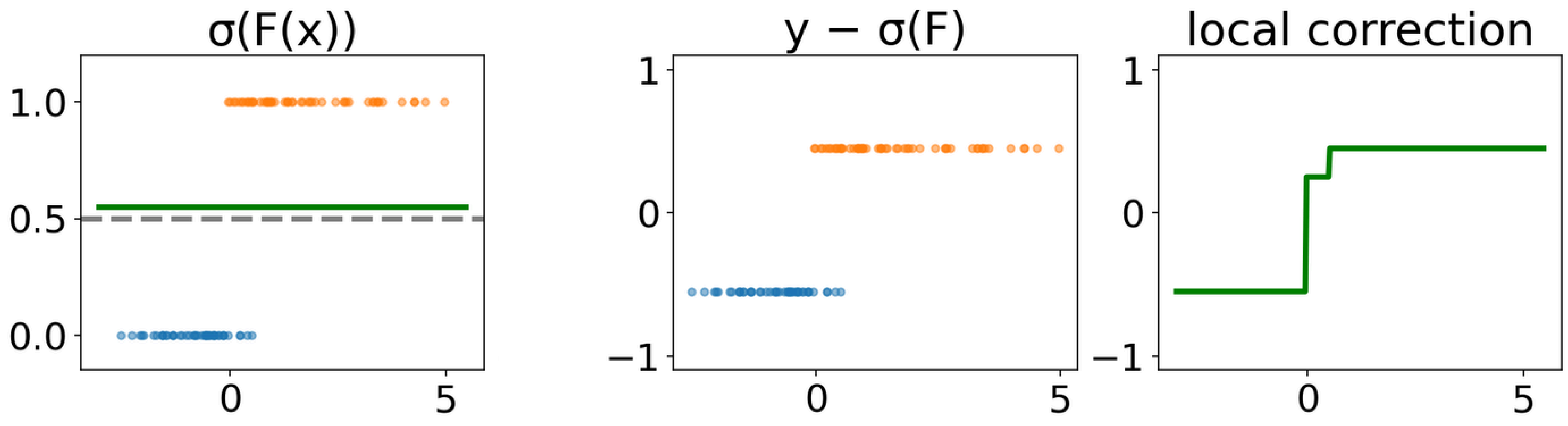


Gradient boosting for classification

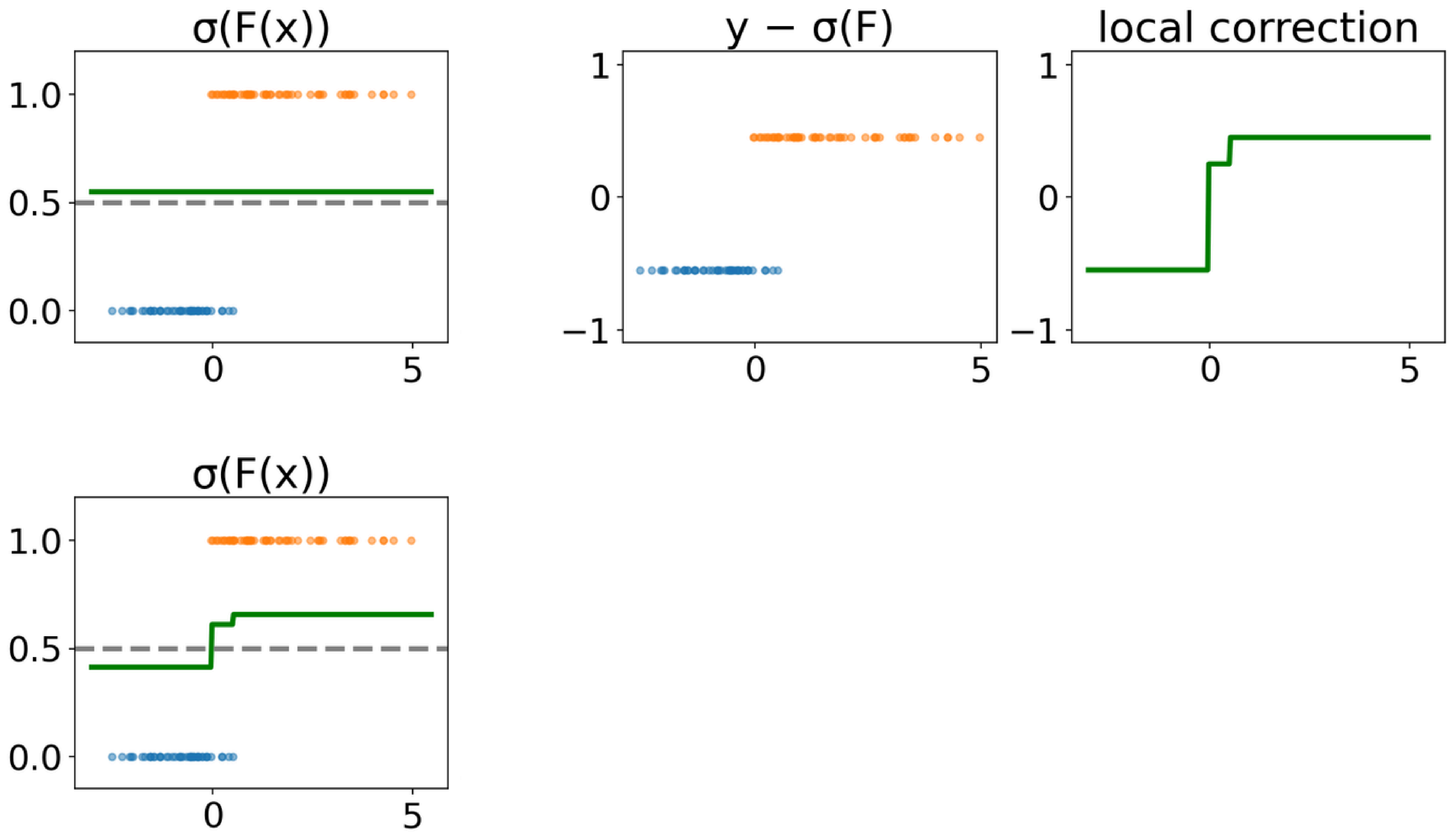


Note: Once again, the residuals can be positive or negative

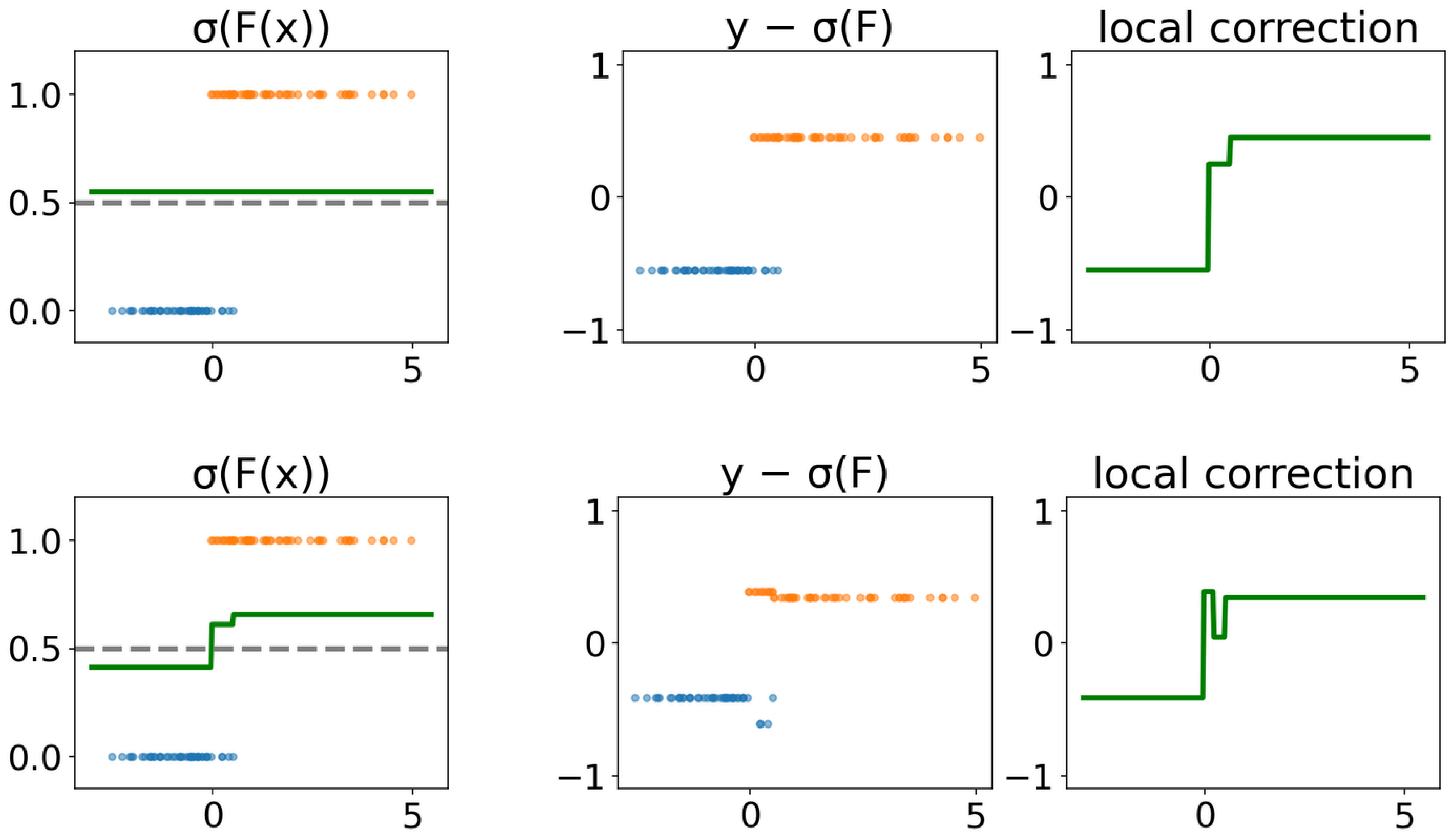
Gradient boosting for classification



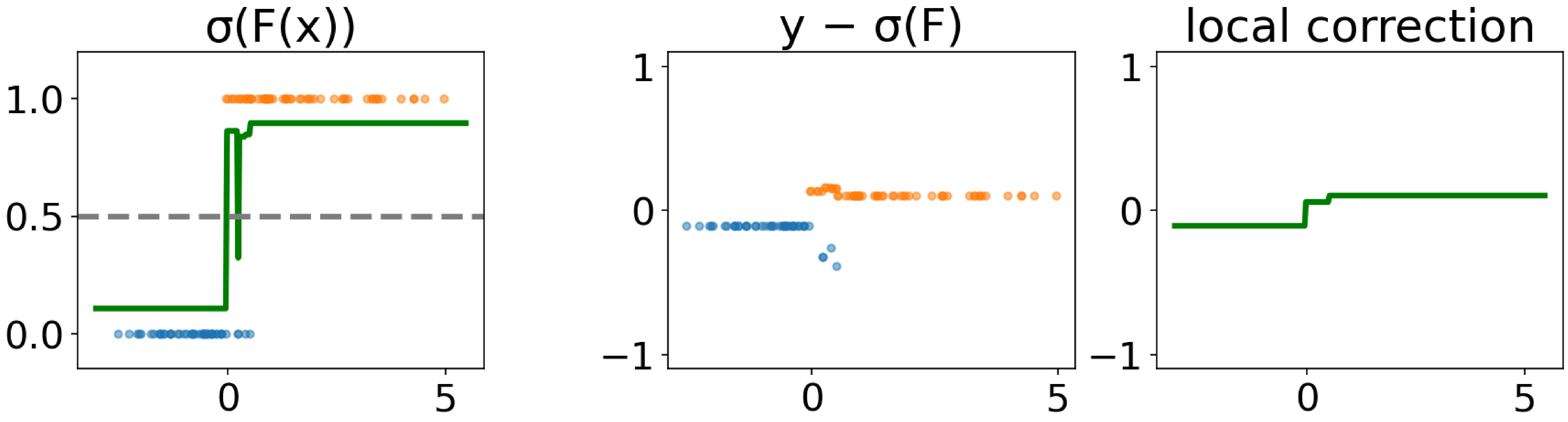
Gradient boosting for classification



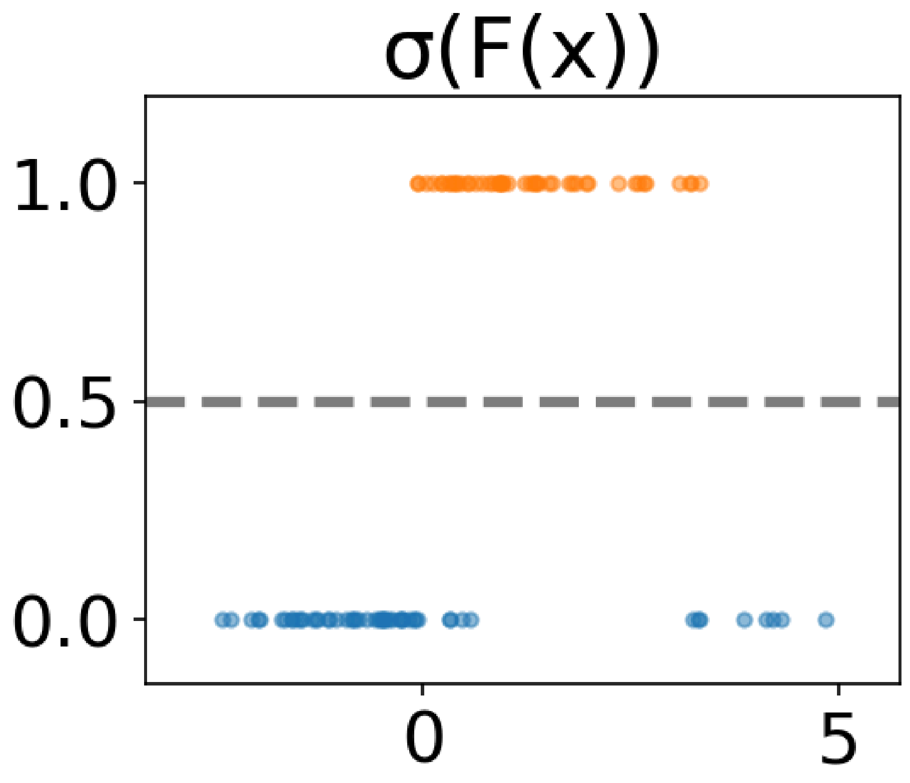
Gradient boosting for classification



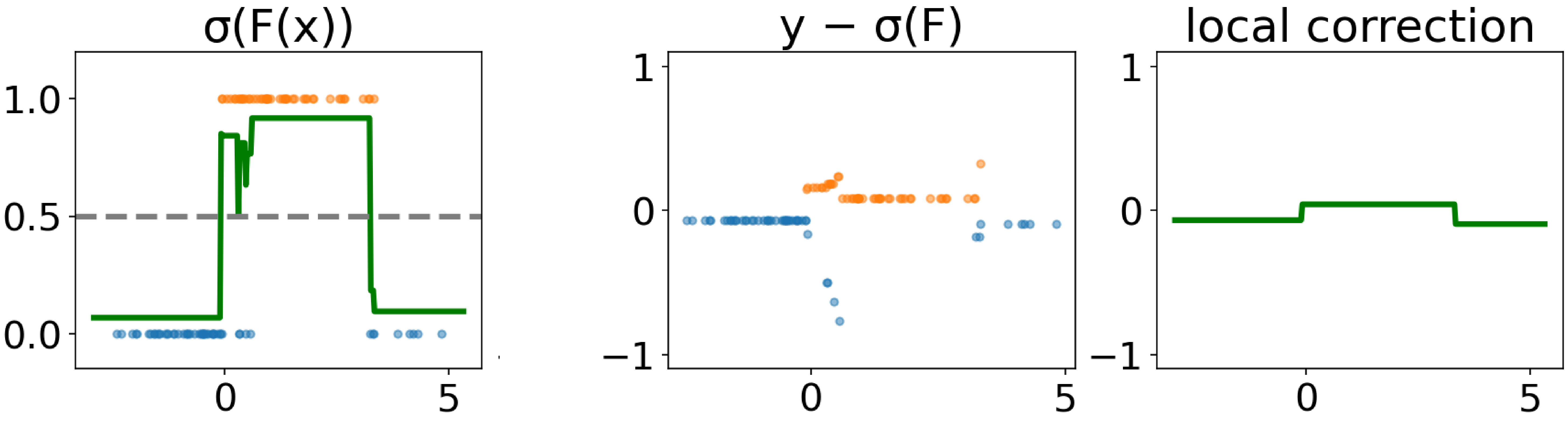
Gradient boosting for classification



Gradient boosting for classification



Gradient boosting for classification



Gradient Boosting and binned features

`sklearn.ensemble.GradientBoostingClassifier`

- Implementation of the traditional (exact) method
- Fine for small data sets
- Too slow for n_samples > 10,000

`sklearn.ensemble.HistGradientBoostingClassifier`

- Discretize numerical features (256 levels)
- Efficient multi core implementation
- Much, much faster when n_samples is large

Take home messages on ensemble models

Bagging	Boosting
fit trees **independently**	fit trees **sequentially**
each **deep tree overfits**	each **shallow tree underfits**
averaging the tree predictions **reduces overfitting**	sequentially adding trees **reduces underfitting**

Gradient boosting tends to perform better than bagging and random forest. Furthermore, shallow trees predict faster.